Ariel Solar & Energy Intelligence (Ariel-SaaS)

COMMERCIAL MASTERCLASS, SREDA PARTNERSHIP ARCHITECTURE & ZERO-CAPEX LAUNCH BLUEPRINT

Executive Summary: This blueprint provides the operational, regulatory, financial, and technical roadmap for deploying Commercial & Industrial (C&I;) rooftop solar across Bangladesh's premier export manufacturing sectors (RMG, Textiles, Jute, Ceramics, and Leather). It outlines the exact mechanisms for launching with zero personal working capital by pairing high-level B2B client relationships with SREDA-enlisted technical subcontractors and Bangladesh Bank green financing.

1. Enterprise Brand Architecture & Venture Positioning

To capture maximum valuation while maintaining institutional authority with factory owners and banks, the venture operates under a dual-engine architecture:

- **Ariel Solar (Commercial/EPC Arm):** The client-facing sovereign clean energy brand. It represents the Commercial, EPC, and SREDA project development arm that signs master turnkey contracts with factory Managing Directors and coordinates banking approvals.
- **Ariel-SaaS™ (Digital Platform & ESCO):** The proprietary digital IoT telemetry, AI battery load-dispatch, and International Renewable Energy Certificate (I-REC) carbon credit tracking platform. Positioned as a climate-tech venture, it qualifies for Startup Bangladesh equity funding (Tk 1–5 Crore) and 4% startup refinancing.
- **Cross-Pollination with ArielMC:** The solar initiative directly feeds the ERP Sourcing Vault. By equipping ArielMC's export factories with solar, finished goods (leather bags, jute carry, ceramics, apparel) can be certified to EU buyers as 100% solar-manufactured, shielding them against the EU Carbon Border Adjustment Mechanism (CBAM).

2. The SREDA Subcontractor Mandate: Why 2 Partners Are Mandatory

Attempting to execute industrial solar installations without SREDA-enlisted engineering partners presents insurmountable legal and financial hurdles. The strategic necessity of locking in two shortlisted engineering firms is defined by three core pillars:

A. Statutory Regulatory Wall & Grid Approval

- **Mandatory SREDA Enlistment:** Under SREDA and Ministry of Power guidelines, a factory cannot receive Net-Metering approval or feed electricity back into the national grid for the guaranteed Tk 10.50/kWh buyback rate unless all single-line diagrams (SLDs), structural load calculations, and synchronization tests are stamped by a SREDA-enlisted firm with OCEI Class-A licensed electrical engineers.
- **Zero Overhead Burden:** Applying for SREDA enlistment independently takes 12 to 18 months, requires auditing past megawatt installations, and demands carrying 3 to 5 certified engineers on permanent payroll (costing Tk 3 to 5 Lakh/month in overhead). Partnering allows Day 1 market entry with zero fixed salary burden.

B. The 'Back-to-Back' Cashflow Protection

- **Client-Funded Milestone Architecture:** The factory client pays a 30% advance upon signing the Master Turnkey Contract (e.g., Tk 77.4 Lakh on a 500 kWp plant). Ariel Solar uses a small fraction of that client advance (e.g., Tk 3.5 Lakh) to fund Milestone 1 (20% mobilization) for the subcontractor to deploy engineers and begin mounting structures.
- **Quality Retention Shield:** A 15% retention fee is held for 60 days post-commissioning and released only upon official utility COD, Performance Ratio verification (PR >= 80%), and certification of zero roof leaks.

C. The 'White-Label' Mandate & Non-Circumvention

- **100% Ariel Solar Branding:** Subcontractor crews and engineers must wear Ariel Solar branded PPE vests and helmets. Under no circumstances may they distribute their own marketing materials or cards.
- **Tk 50 Lakh Legal Penalty:** The Memorandum of Understanding (MoU) binds the subcontractor under an ironclad non-circumvention clause backed by an immediate liquidated damages penalty of Tk 50,00,000 (Tk 50 Lakh) if they

attempt to bypass Ariel Solar to deal directly with the factory owner.

D. Commercial Leverage: Why TWO Shortlisted Firms?

- **Price Discovery & Margin Protection:** Firm A quotes Tk 3,800/kWp while Firm B quotes Tk 3,200/kWp. Using Firm B as a benchmark allows negotiating Firm A down to Tk 3,300/kWp. On a 500 kWp plant, a Tk 500/kWp reduction equals Tk 2,50,000 in immediate cash savings for Ariel Solar.
- **Regional Redundancy:** If one partner is fully committed on a project in Gazipur, the second partner can deploy immediately to Savar, Narayanganj, or Chittagong, preventing deal bottlenecks.

3. Master Launch Checklist: Everything Needed to Start

Category 1: Legal & Corporate Infrastructure

- **[ ] Corporate Entity:** Incorporate Ariel Solar Limited or establish it as an executive industrial division under Ariel Marketing & Communications Limited.
- **[ ] Trade License Scope:** Renewable Energy Services, Solar EPC Contracting, Electrical Engineering Consulting, and Hardware Import.
- **[ ] Tax & Regulatory:** Corporate TIN, 13-digit BIN, and SREDA digital vendor tracking profile.
- **[ ] Bank Accounts:** Dual-currency corporate account for BDT project advances and USD ERQ import facilitation.
- **[ ] Standard Legal Suite:** Master EPC Turnkey Contract, White-Label Subcontractor MoU (with Tk 50L penalty), and Factory NDA.

Category 2: Technical & Engineering Partnerships

- **[ ] 2 SREDA Partners:** Execute white-label MoUs with two Dhaka/Chittagong firms having OCEI Class-A licenses.
- **[ ] Satellite Audit Tool:** Google Earth Pro desktop for remote area and obstacle measurement.
- **[ ] PVsyst Simulations:** Partner engineer runs 8760-hour hourly generation simulation and single-line diagrams (SLD).
- **[ ] Drone Survey Access:** Freelance RTK drone operator for 3D roof mapping and thermal baseline audits.

Category 3: Banking & Green Refinancing Partnerships

- **[ ] Bank SFD Tie-Ups:** Formal introductions with Sustainable Finance heads at BRAC Bank, Eastern Bank (EBL), and City Bank.
- **[ ] IDCOL Sourcing Desk:** Familiarity with the 10-year term, 80% debt, 1-year grace period concessionary loan facility.
- **[ ] Refinancing Packages:** Structuring loan packages where monthly bill savings exceed loan installments by Tk 4+ Lakh/month.

Category 4: Hardware Supply Chain & Duty Exemptions

- **[ ] Tier-1 Solar Modules:** Direct supply channels for Jinko (Tiger Neo), Trina (Vertex N), and LONGi (Hi-MO 7). Concessional 1% duty under H.S. 8541.43.00.
- **[ ] Tier-1 Inverters:** Authorized channels for Huawei (SUN2000) and Sungrow (SG110CX). Concessional 1% duty under H.S. 8504.40.90.
- **[ ] Balance of System:** Anodized aluminum rail clamps for sandwich panels (zero roof piercing) and 1500V UV solar cables (BRB/Lapp).

Category 5: Sales Engine & Marketing Collateral

- **[ ] 4-Page Bespoke Dossier:** Executive PDF format with 3D satellite overlay, bill comparison, and green loan waterfall.
- **[ ] Digital Solar Portal:** Interactive sizing and ROI calculator deployed on solar.arielmc.com.
- **[ ] Pilot Prospect List:** 10 friendly factory owners in Gazipur, Savar, and Narayanganj selected from existing ArielMC relationships.

4. Benchmark Financial & Engineering Model (500 kWp Plant)

Financial & Operational Metric	Benchmark Figure (Dhaka/Gazipur)
System Capacity	500 kWp N-Type TOPCon (Bifacial)
Required Net Roof Space	3,400 sq. meters (~36,500 sq. ft.)
Turnkey Client Contract Price	**Tk 2,58,00,000 (Tk 2.58 Crore)**
Hardware & BoS Cost	Tk 1,92,00,000 (Tk 1.92 Crore)
Subcontracted Labor (Tk 3,400/kWp)	Tk 17,00,000 (Tk 17 Lakh)
Ariel Solar Gross Margin	**Tk 49,00,000 (19.0% Blended Margin)**
Annual Solar Generation	725,000 kWh / Year
Annual Electricity Bill Savings	**Tk 1,05,12,500 / Year (Tk 8.76 Lakh/Mo)**
Monthly Green Bank Loan EMI (4.5%)	Tk 3,85,000 / Month
Net Monthly Cash Gain for Factory	**+Tk 4,91,000 / Month (Net Positive)**
Simple Client Payback Period	3.4 to 3.8 Years
Annual Carbon Abatement	435 Metric Tons CO2e / Year (I-RECs)

5. 14-Day Tactical Execution Roadmap

- **Days 1–3 (Digital Tooling):** Finalize the interactive Solar Sizing & ROI Calculator on the portal; format the 4-page Bespoke Dossier template.
- **Days 4–7 (Subcontractor Lock-In):** Issue the White-Label MoU to 2 shortlisted SREDA engineering firms; negotiate installation labor to Tk 3,200–3,400/kWp and secure non-circumvention signatures.
- **Days 8–10 (Pilot Dossier Generation):** Select 1 friendly export factory owner in Gazipur or Savar. Measure their roof on Google Earth Pro, run the numbers, and compile their custom 4-page proposal.
- **Days 11–14 (The Pitch & Closing):** Host an executive meeting with the factory MD/CFO. Pitch the zero-capital, cashflow-positive green loan structure and sign the 1-Day Free On-Site Drone & Structural Audit Agreement.